Computer Vision

T2 Learning CNNs

University Master's Degree in
Machine Learning

upna

Universidad Pública de Navarra
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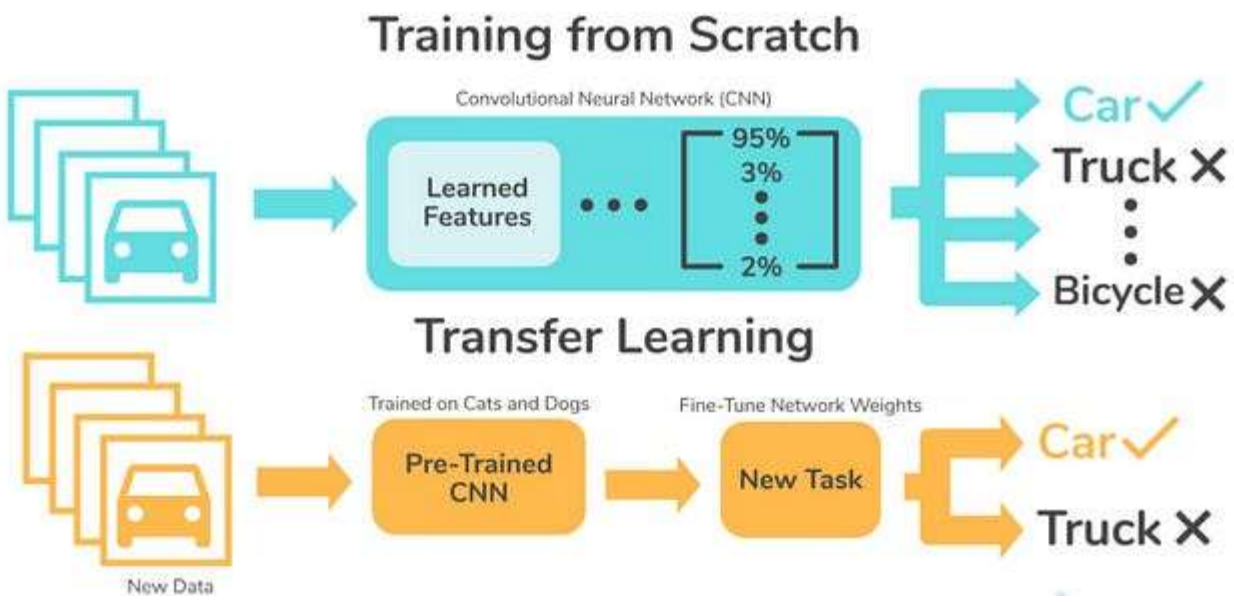
Transfer Learning

- Training a CNN from scratch requires:
  - Huge labeled datasets
  - Weeks or months of training
  - Powerful GPUs
- Not always feasible for specific tasks
  - Example: Suppose you want to build a CNN to detect rare diseases in X-ray images
    - To train from scratch, you might need millions of labeled medical images
    - But in reality:
      - Only a few thousand images exist
      - Labeling requires specialized radiologists (time + cost)
      - Training such a large model would need weeks on expensive GPUs

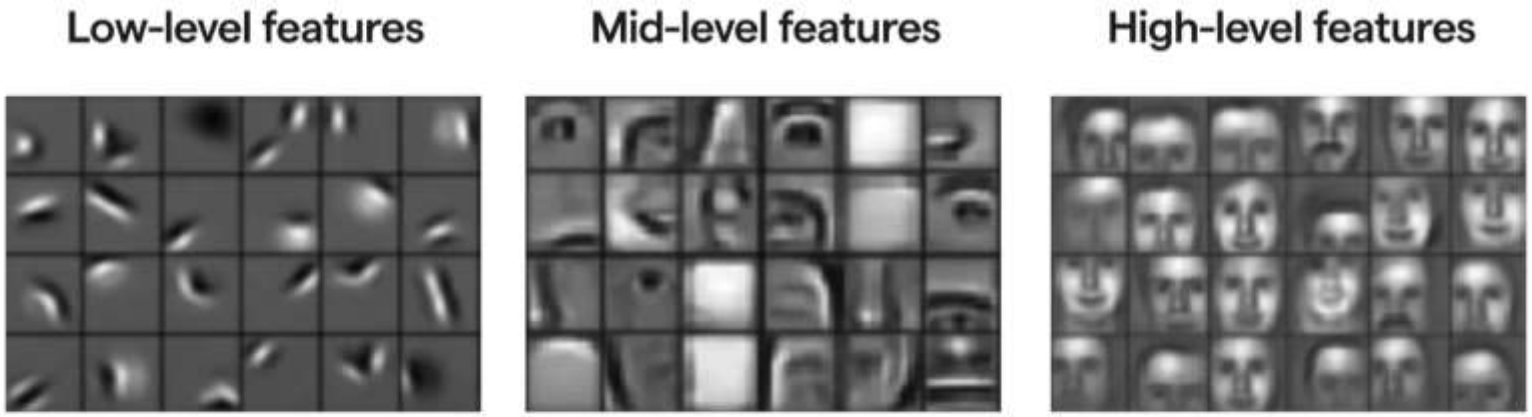
Transfer Learning

- What is Transfer Learning?
  - Using a pretrained neural network as a starting point
  - Transfer of knowledge (weights & features) from one task/domain to another
  - Train on a smaller dataset for your new task (fine-tunning)



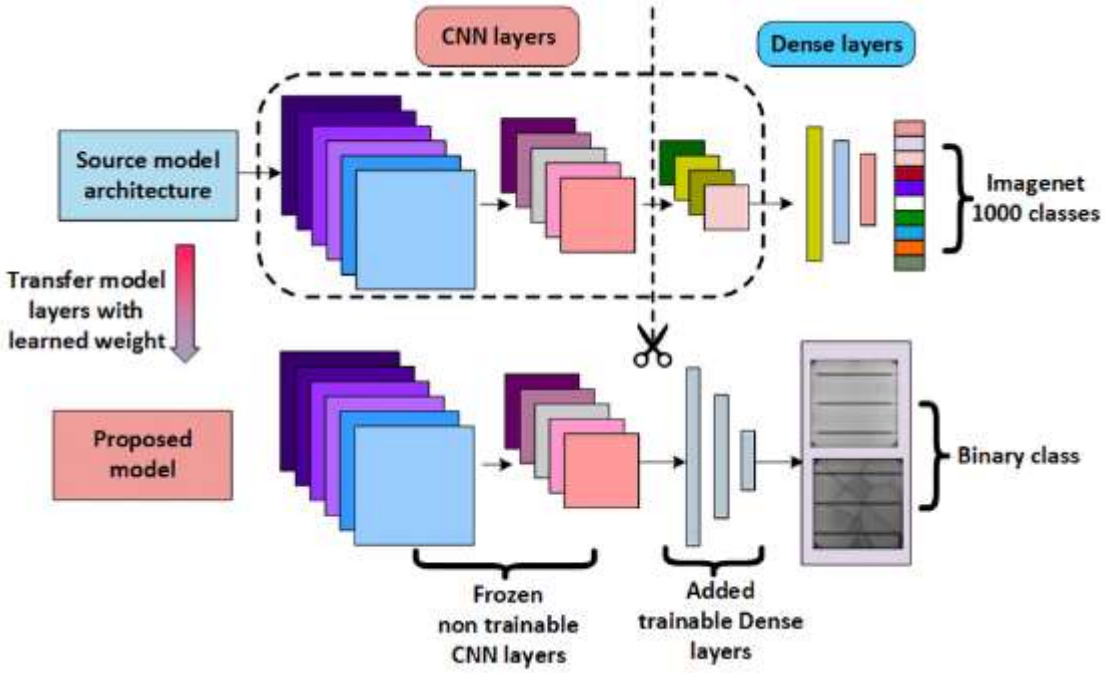
Transfer Learning

- Why Transfer Learning works?
  - Pretrained networks already learn generic features:
    - Early layers: edges, textures, colors
    - Mid layers: shapes, patterns
    - Later layers: task-specific features
  - You only need to fine-tune or adapt for your problem



Transfer Learning

- Benefits of Transfer Learning
  - Reduces Training Time
  - Requires less labeled data
  - Less prone to overfit
  - Achieves higher performance
  - Reuses state-of-the-art research

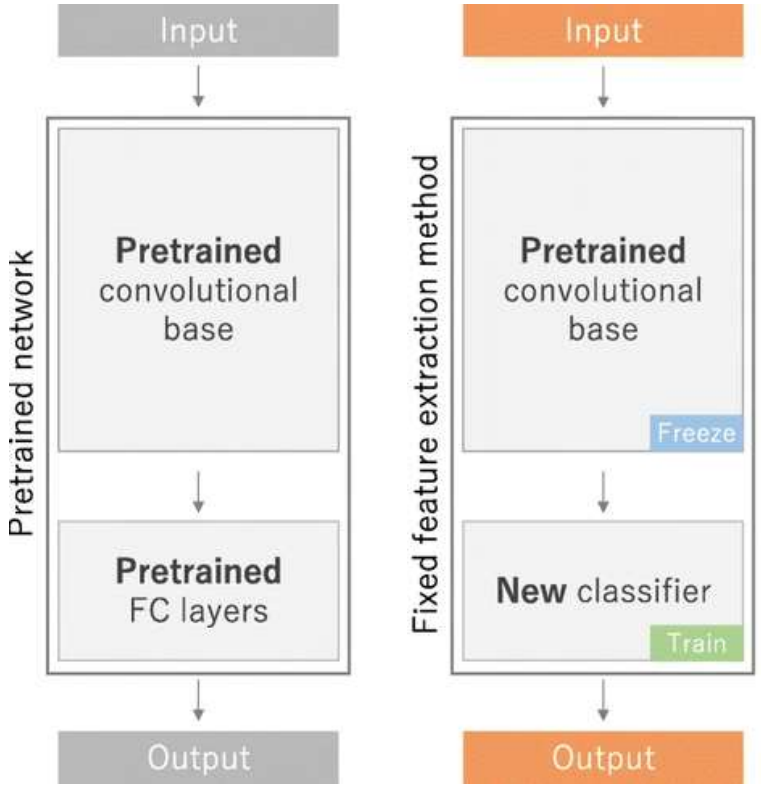


Transfer Learning

- Transfer Learning approaches
  - Feature extraction
    - Freeze early layers, replace top layers
  - Fine-tunning
    - Fine-tune some layers

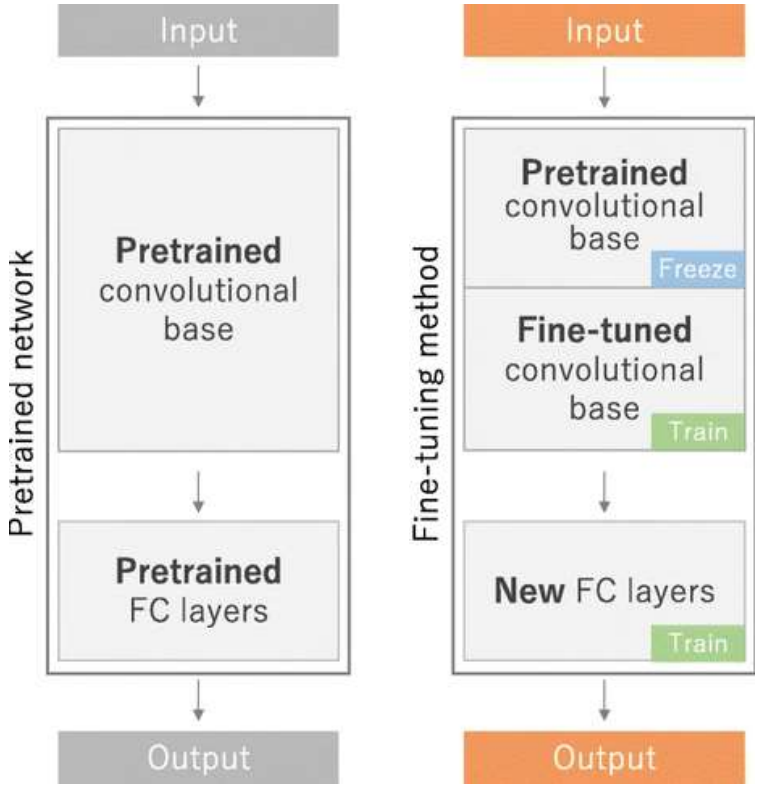
Transfer Learning

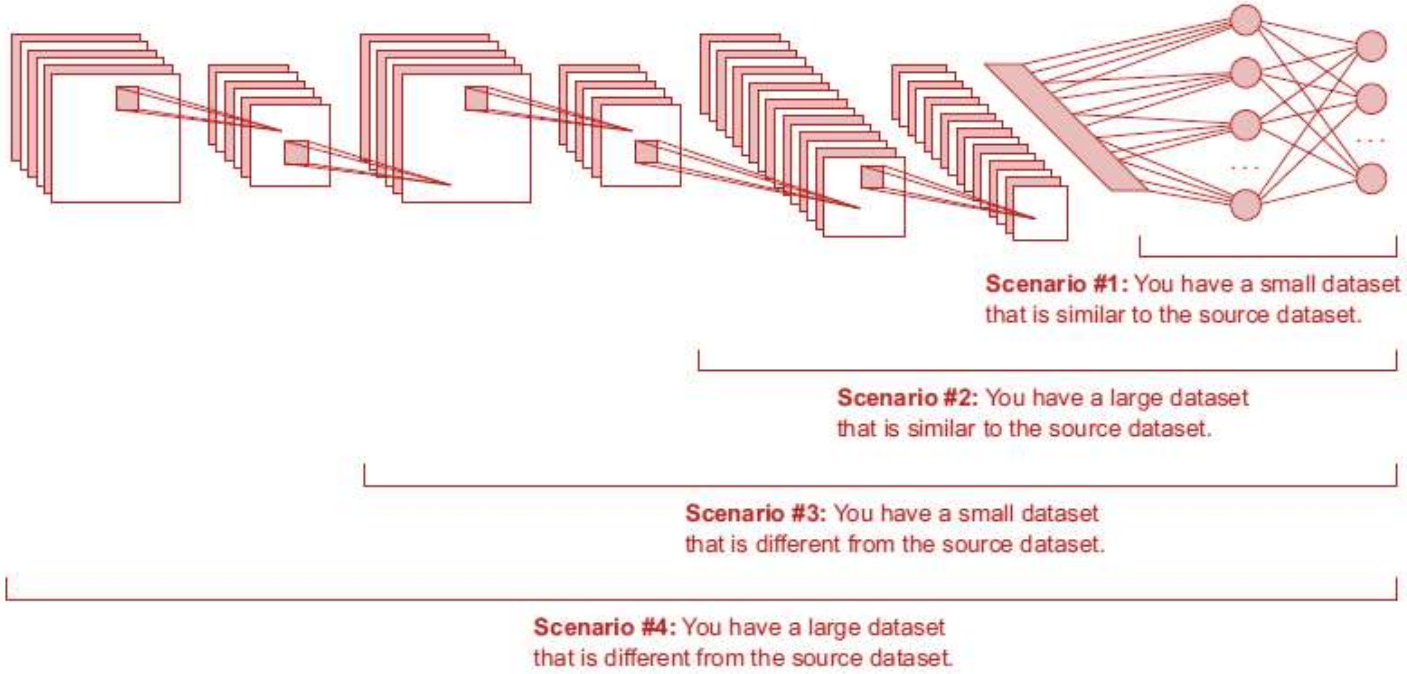
- Transfer Learning approaches
  - Feature extraction
    - Use the pre-trained model to extract features
    - Freeze earlier layers as they capture generic features
    - Fine-tune only the top layers



Transfer Learning

- Transfer Learning approaches
  - Fine-tunning
    - Unfreeze some or all layers of the pre-trained model
      - The more the data you have the more layers you can unfreeze
      - Popular approach: gradual unfreezing
    - Train the network end-to-end
      - Use small learning rates to avoid overwriting pre-trained weights
      - It is possible to apply gradually increasing layer-wise learning rates



Transfer Learning

Scenario	Size of the target data	Similarity of the original and new datasets	Approach
1	Small	Similar	Pretrained network as a feature extractor
2	Large	Similar	Fine-tune through the full network
3	Small	Very different	Fine-tune from activations earlier in the network
4	Large	Very different	Fine-tune through the entire network